JAE YOUNG JANG

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EDUCATION

Ph.D.	**Korea Advanced Institute of Science and Technology** *Department of Civil and Environmental Engineering* *Graduate School of Metaverse*	**Mar. 2025 – Present** Daejeon, Korea
M.S.	**Korea Advanced Institute of Science and Technology** *Graduate School of Metaverse* Thesis: *Complex Human Activity Recognition in Residential Spaces with Sector-Level Approach* (Advisor: Seung Hyun Cha)	**Mar. 2023 – Feb. 2025** Daejeon, Korea
B.S.	**Seoul National University of Science and Technology** ***GPA: 4.18/4.5 (7th place out of 150)*** ***Major GPA: 4.39/4.5*** *Department of Computer Science and Engineering*	**Mar. 2017 – Feb. 2023** Seoul, Korea

RESEARCH PROFILE

My research focuses on **context-aware smart home automation**, integrating **contextual information** about users and their environments to enable both **reactive and proactive control.**

RESEARCH FUNDING

National Research Foundation of Korea (NRF) *2025–2027*
Ph.D. Student Research Encouragement Grant

- **Role: Principal Investigator**
- Title: Sector-level Spatial Segmentation and Multimodal Sensor Fusion for Complex Human Activity Recognition in Residential Environment

PUBLICATIONS

- **International Journals**
  - **Jang, J. Y.**, & Cha, S. H. (2026). Context-integrated complex activity recognition with sector-level location encoding in residential environments. *Advanced Engineering Informatics*, *76*, 105043. **[SCIE, Q1, 2.0%]**
  - Seol, H., Yang, J., **Jang, J. Y.**, Choi, D., Han, G., & Cha, S. H. (2026). Development of a Physical Security Risk Matrix Integrating Heterogeneous Sensor Data and Spatiotemporal Context: Expert-Derived Parameters and Scenario-Based Validation. *Architectural Research*, *28*(1), 28.
  - Song, Y., **Jang, J. Y.**, Han, G., & Cha, S. H., A Computational Framework for Integrating Activity-Based Space Utilization Optimization and Office Layout Generation. *Journal of Computational Design and Engineering* (Under Review).
- **International Conferences**
  - Song, Y., **Jang, J. Y.**, Eum, Y., & CHA, S. H. (2024). Space Utilization Based Framework For Designing Office Buildings For Efficient Space Use. In *International Conference of Smart and Sustainable Built Environment (SASBE 2024)*. Auckland University of Technology.

- Song, Y., **Jang, J. Y.**, Eum, Y., & Cha, S. H. (2024, November). A Conceptual Framework of Optimized Office Floor Plan Generation Based on Space Utilization Prediction. In *The International Conference of Sustainable Development and Smart Built Environments* (pp. 34–43). Singapore: Springer Nature Singapore.

- **International Posters**
  - Bae, S. A., Eum, Y., Kim, H. W., Kim, Y., Jeon, D., **Jang, J. Y.**, & Cha, S. H. (2025, October). Adaptive Virtual Reality Space Design Based on Real-Time User Emotion. In *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)* (pp. 63–66). IEEE.
- **Domestic Journals**
  - Song, Y., **Jang, J. Y.**, & Cha, S. H. (2024). Development of Methodology for Automated Office Room Generation Based on Space Utilization. *Journal of KIBIM*, *14*(3), 1–12.
- **Domestic Conferences**
  - Yang, J., **Jang, J. Y.**, Seol, H., Kwon, D., Choi, D., Cheon, H.-C., Han, G., & Cha, S. H. (2026). An AI-City Integrated Physical Security Monitoring Framework: VLM Anomaly Detection, Heterogeneous Data Fusion-Based Risk Inference, and LLM-Based SOP Automation. *Proceedings of the 2026 Summer Conference of the Society for Computational Design and Engineering*.
  - **Jang, J. Y.**, Kim, H., & Cha, S. H. (2024). Occupant Activity Recognition in Smart Homes - The Need for Expanded Research on Sensor-Based Human Activity Recognition. *Proceedings of the Spring Annual Conference of the Architectural Institute of Korea*, *44*(1), 482–485.
  - **Jang, J. Y.**, Song, Y., & Cha, S. H. (2023). Automatic floorplan generation considering energy efficiency - Focused on the Energy Performance Prediction. *Proceedings of the Autumn Annual Conference of the Architectural Institute of Korea*, *43*(2), 501–504.
- **Domestic Posters**
  - Jung, K., Park, J., **Jang, J.**, Kim, H. I., & Woo, W. (2024). TeleAssiBot: Dual Secretary Assistive Robot. *Proceedings of the HCI Korea Conference*, 1004.

RESEARCH PROJECTS

- Sector-level Spatial Segmentation and Multimodal Sensor Fusion for Complex Human Activity Recognition in Residential Environment, PI, National Research Foundation 2025–2027
- Development and Commercialization of AI-based Next-Generation Physical Security Integrated Monitoring System, Researcher, Korea Internet & Security Agency 2026
- Development of Core Technologies for the Design, Construction, and Lifecycle Management of Robot-Adaptive Buildings, Researcher, Korea Agency for Infrastructure Technology Advancement 2025–2029
- AI-Based Responsive-Transformative Spatial Technology for Future Space Innovation, Researcher, National Research Foundation 2025–2028
- AR-AI-Cognition-Architectural Research for Transdimensional Interaction, Researcher, National Research Foundation 2024–2025
- Real-time XR Interface Technology for Environmental Adaptation, Researcher, Institute for Information & Communications Technology Planning & Evaluation 2024–2025
- Establishment of an International Network for Generative Agent Simulation-based Architecture Design Standard, Researcher, National Research Foundation 2024–2025
- Development of Pre-Construction Spatial Evaluation System based on Multimodal Virtual Reality, Researcher, National Research Foundation 2023–2028
- Space Design Framework for the Metaverse, Researcher, National Research Foundation 2022–2025
- Development of IoT/AI based building envelope platform responsive to air-borne disease (COVID-19) and fire disasters, Researcher, National Research Foundation 2022–2027
- Development of Artificial Intelligence Based Architectural Design Automation Technologies, Researcher, Korea Agency for Infrastructure Technology Advancement 2021–2025

AWARDS

- **2024 HCI Korea Best Poster Award**
- **2023 AIK (Architectural Institute of Korea) Best Paper Presentation Award**
- **2021 Big Data Platform Utilization Idea Contest (Encouragement Prize)**